Banking Analytics: Key Insights, Recommendations & Challenges

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Scope and Definitions

The analysis covers 5,050 customers, 5,100 accounts, 20,000 transactions (12 May 2022 to 11 May 2025), 2,600 loans, 4,000 cards and 3,100 support calls (11 May 2024 to 11 May 2025). The model is built around the Customers table, with a Calendar table related to join date.

Because of that relationship, the dashboard Year filter means customer join year (a cohort). Time trends use each table's own date column instead.

Term	Definition
Reference Date	11 May 2025, the last transaction in the data. Used instead of today for every activity measure.
Active customer	Last transaction within 90 days of the Reference Date.
Cooling customer	Last transaction 90 days to 6 months before the Reference Date.
Churn Risk customer	Last transaction more than 6 months before the Reference Date.
No Account / No Transactions	Customer with no account at all / customer with an account but no transactions.
Balance at Risk	Total balance held by Churn Risk customers.
Dormant account	Last transaction more than 6 months before the Reference Date. Accounts with no transaction at all are classed separately as Never Used.
High-value transaction	Amount of $9,000 or more (about the top 10% of transactions).
Net Flow	Total deposits minus total withdrawals.

Key Insights

The bank lends far more than it holds, and most of its customers, accounts and loans sit with people who are inactive or have no account.

1. **Lending is 2.5 times balances.** Loans total $644.3M against balances of $254.7M, a Loan-to-Balance Ratio of 253%.

2. **Only 24% of customers are active.** Of 5,050 customers, 1,206 are Active (23.9%), 686 Cooling (13.6%) and 1,252 Churn Risk (24.8%). The other 1,906 (37.7%) have no account (1,851) or no transactions (55). The mix is similar in every join year (Churn Risk 21% to 28%), so tenure does not explain inactivity.

3. **A third of balances is exposed.** Churn Risk customers hold $86.1M, 33.8% of all balances.

4. **Half of accounts are dormant.** 2,516 accounts (49.3%) holding $128.4M (50.4% of balances) are dormant, and another 181 accounts ($9.4M) have never been used. 699 dormant or never-used accounts hold $75,000 or more ($61.3M). Dormancy is 48% to 51% in every account type and 45% to 53% across opening years, so neither type nor age explains it.

5. **Transaction activity is steady and balanced.** About 6,650 transactions ($33M) a year in 2023 and 2024, split almost equally across deposits, payments, transfers and withdrawals. Net Flow is +$0.6M (deposits $24.98M, withdrawals $24.38M).

6. **A small set of transactions and accounts needs review.** 2,042 transactions (10.2%) are $9,000 or more ($19.39M), and 38 accounts have 10 or more transactions against an average of 4.1.

7. **Loan risk is concentrated in inactive customers.** Only 23.5% of loan value ($151.5M) belongs to Active customers. 36.8% ($237.3M) belongs to customers with no account and 26.0% ($167.4M) to Churn Risk customers.

8. **A maturity wall is coming while new lending has slowed.** 996 loans ($247.5M, 38% of the book) mature within 24 months, 509 of them ($126.8M) within 12 months. New loans fell 56% in 2024 (255 against 585 in 2023). Business loans are only 1.1% of loans but carry the highest average rate, 9.0% against 7.4% to 7.7% for other types.

9. **Support resolves only half of calls.** 1,571 of 3,100 calls (50.7%) are unresolved, and no issue type is above 52% resolved. Card Issue is weakest (46.5%) and Transaction Dispute is the most frequent (803 calls). 203 customers have two or more unresolved calls, and 36% of calls come from customers with no account.

Business Recommendations

The actions below are ordered by the size of the exposure they address.

1. **Reactivate Churn Risk and Cooling customers, starting with high-value dormant accounts.** Contact the 699 dormant or never-used accounts holding $75,000 or more ($61.3M) first, then the 1,252 Churn Risk customers ($86.1M in balances) with a personal call or relationship-manager offer. Track the Active share (23.9%) and dormant share (49.3%) monthly.

2. **Run an account-opening drive for customers with no account.** 1,851 customers have no account, yet many hold loans (36.8% of loan value). Convert borrowers first by linking their existing loans to a real account.

3. **Protect the loan book through renewal and early monitoring.** Start renewal or refinancing conversations for the 509 loans ($126.8M) maturing within 12 months. Monitor the $167.4M of loans held by Churn Risk customers as a watch list, and review the 56% fall in new lending in 2024.

4. **Review funding and set a Loan-to-Balance limit.** At 253%, lending is far above balances. Agree an internal threshold, and grow selectively in higher-rate segments such as Business loans (9.0% average rate) only where customers are active.

5. **Improve customer support follow-up.** Contact the 203 customers with two or more unresolved calls, find the root causes behind Card Issue (46.5% resolved) and Account Access, and set a resolution-rate target above 80%. Start recording resolution date, agent, priority and satisfaction score so speed and quality can be measured.

6. **Monitor high-value and high-frequency activity.** Review the 2,042 transactions of $9,000 or more and the 38 accounts with 10 or more transactions against normal behavior, and set alerts for repeated large

transactions.

From Insight to Action

Each problem found in the data is matched to the action that solves it, the owner, and the dashboard measure that shows whether it worked.

Problem (insight)	What to do	Team	Success measure
Only 23.9% of customers are active; 24.8% are Churn Risk	Run a win-back campaign: personal calls and offers to Churn Risk and Cooling customers, with the highest balances first	Retention / Relationship managers	Active % up, Churn Risk % down each month
49.3% of accounts are dormant; 699 high-value dormant or never-used ones hold $61.3M	Contact the owners of high-value dormant accounts, offer an incentive for a first transaction, and close or merge accounts that stay unused	Retention / Operations	Dormant Accounts and Dormant Balance down
1,851 customers have no account but hold loans	Open an account for every borrower and link their loans to it	Branches / Onboarding	No Account % down, Account Holders % up
Loans are 253% of balances	Set a Loan-to-Balance limit, raise deposits with savings offers, and slow lending to inactive customers	Risk / Finance	Loan-to-Balance Ratio trending down toward the agreed limit
36.8% of loan value belongs to customers with no account and 26.0% to Churn Risk customers	Put these loans on a watch list, with early payment reminders and account-opening offers	Credit risk	Loans at Risk % down
509 loans ($126.8M) mature within 12 months and new lending fell 56% in 2024	Offer renewal or refinancing 6 to 12 months before maturity, and plan new lending for active customers	Lending	Loans Maturing 12M renewed, new loans per year up
Only 49.3% of support calls are resolved; Card Issue is weakest at 46.5%	Add follow-up for open calls, fix the root cause of Card Issue and Account Access calls, and train agents on those topics	Customer support	Resolution Rate % up toward 80%
203 customers have 2 or more unresolved calls	Escalate these customers to a senior agent within a set time	Customer support	Repeat Callers down
2,042 transactions of $9,000 or more and 38 accounts with 10+ transactions	Review them against normal behavior and set automatic alerts for repeated large transactions	Compliance / Fraud	High-Value Transactions reviewed, alerts raised

Challenges Faced During the Project

1. **Calendar relationship.** The Calendar table is related only to Customers[JoinDate], so the Year slicer filters customers by join year, not by transaction date. Time trends therefore use each table's own date

columns (transaction month, call month, loan start year), with the Year slicer kept as a customer cohort filter.

2. **Data ends in May 2025.** TODAY() made activity measures return zero or meaningless values, so a Reference Date (last transaction date, 11 May 2025) defines Active, Cooling, Churn Risk and Dormant.

3. **Partial years and months.** 2019 to 2025 are not all complete, and May 2022 and May 2025 are partial months. The Overview monthly trend excludes them so it does not show false drops, while the Support monthly chart still includes May 2024 and May 2025.

4. **Missing fields for risk analysis.** The data has no default or overdue status, income, resolution date, agent, priority or satisfaction score. Inactivity was used as a proxy for risk, and Loan-to-Income and resolution-time analysis could not be done.

5. **Data consistency issues.** Many customers with loans have no account, support issue types do not match the customer's products, some accounts and loans carry 2026 dates, and one customer category (Army) holds 531 customers. These were kept and reported as findings, not removed. Army was left out of the Top 10 state charts because it dominated them.

6. **Near-uniform distributions.** Categories such as transaction type, season, loan type and issue type are almost evenly split, so differences between segments are small. Insights focus on customer status, dormancy and maturity, where the data shows clear gaps.

7. **Technical issues.** Automatic date hierarchies blocked date filtering, and measures needed REMOVEFILTERS and careful row-context handling so counts did not change when a slicer was applied. Verifying every measure against the raw data caught these before the final dashboard.